Real-time deep learning-based photon segment dose calculation

Fan Xiao[1], Niklas Wahl[2,3], Claus Belka[1,4,5], George Dedes[6,1], Christopher Kurz[1], Guillaume Landry[1,5]

[1] Department of Radiation Oncology, LMU University Hospital, LMU Munich, [2] Department of Medical Physics in Radiation Oncology, German Cancer Research Center (DKFZ), [3] National Center for Radiation Oncology (NCRO), Heidelberg Institute for Radiation Oncology (HIRO), [4] German Cancer Consortium (DKTK), [5] Bavarian Cancer Research Center (BZKF), [6] Department of Medical Physics, Faculty of Physics, Ludwig-Maximilians-Universität München

Department of Radiation Oncology, LMU University Hospital, LMU Munich

Purpose

- Deep learning (DL) enables sub-second dose calculation, critical for online adaptive radiotherapy re-optimization
- Two coordinate paradigms compared: Beam's Eye View (BEV) vs. patient-coordinate
- Representative DL models evaluated on a unified dataset and processing pipeline
- Two novel lightweight DL models introduced for efficient photon segment dose calculation

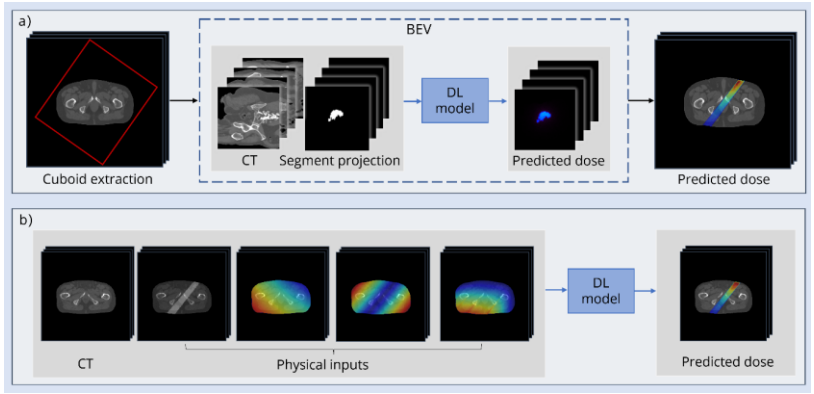


Figure 1. DL dose calculation: (a) BEV coordinate modeling, (b) Patient coordinate modeling (CT+physical inputs)

Methods

- CT and volumetric modulated arc therapy (VMAT) plans from 24 prostate cancer patients
- 5940/540/3053 Monte Carlo (MC) segment doses for training (11 patients), validation (3), and testing (10)
- 4 BEV models: CNN-ConvLSTM, CNN-Mamba, DoTA[1], C3D[2]
- 1 patient-coordinate model: DeepDose-C3D[3]
- Accuracy: local γ (2%/3mm) for segment doses, local γ (1%/3mm) + dose volume histogram (DVH) for plan doses

Results

- All 5 models: mean γ (2%/3mm) ≥ 91.0% for segment doses; patient-coordinate model performed worse in segment accuracy
- All 5 models: mean local γ (1%/3mm) ≥ 99.0% for plan doses
- Segment calculation time: CNN-ConvLSTM 79 ms, CNN-Mamba 67 ms, DoTA 298 ms, C3D 490 ms, DeepDose-C3D 356 ms

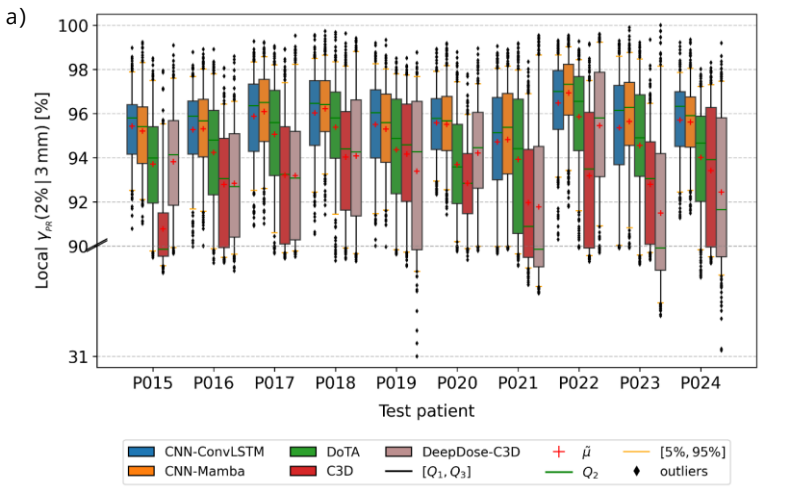


Figure 2. (a) Patient specific boxplot for local ypr of predicted segment dose from five models in test dataset, (b) DVH comparison between MC plan dose and five models' plan doses for a test patient

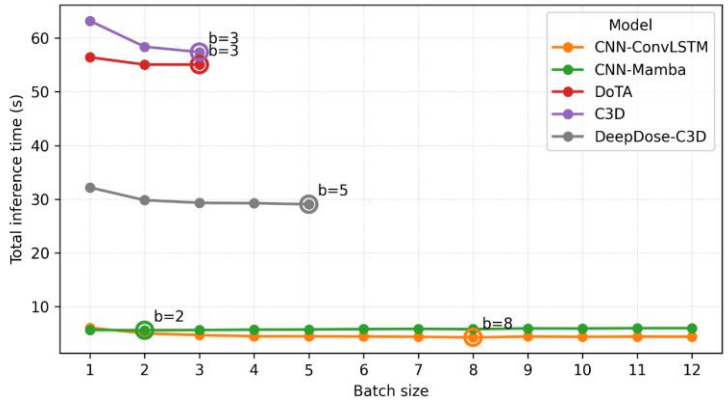


Figure 3. Total inference time vs. batch size for five models on a single test plan (254 segments) using an RTX A6000 GPU

Conclusion

- BEV and patient-coordinate DL methods both achieve accurate plan dose calculation
- BEV-based models show more robust segment-level performance
- CNN-ConvLSTM & CNN-Mamba: comparable accuracy at faster speed and lower computational memory cost

Acknowledgments

The work of Fan Xiao was supported by China Scholarship Council (No.202308440107), and this project was supported by Deutsche Forschungsgemeinschaft (No.469106425).

GITHUB LINK

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